WRITTEN REPORT

Java OOP Laboratory Exercise: Campus Café Ordering System

Names: Jan Clarence Valencia; Maceda Dennis Valerian

Section: D

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Exercise Title: Campus Café Ordering System

I. Introduction

The Campus Café Ordering System is a Java program designed to process customer orders in a university café. It allows customers to order up to three menu items, including beverages, pastries, and meals. The program uses Object-Oriented Programming (OOP) concepts to organize the code, calculate prices, and display the customer's order and total payment.

II. Explanation of OOP Concepts

1. Classes and Objects
Classes were used to organize the different parts of the ordering system, such as customers, menu items, and orders. Objects were created from these classes to represent actual customers and food items.

2. Constructors
Constructors were used to initialize the attributes of objects when they were created. They help assign initial values to customers and menu items.

3. Encapsulation
Encapsulation was applied by making class attributes private and providing public getter and setter methods. This protects the data and allows controlled access to the information.

4. Inheritance
Inheritance was used to allow specific menu item classes, such as Beverage, Pastry, and Meal, to inherit common attributes and methods from a parent class. This reduces repeated code and improves organization.

5. Polymorphism
Polymorphism was used through method overriding to allow different menu item types to calculate their prices differently. Each item can have its own implementation of the price calculation method.

6. Abstraction
Abstraction was applied through an abstract parent class that defines common methods for menu items. It hides unnecessary implementation details and allows subclasses to provide their own behavior.

7. Conditional Statements
Conditional statements were used to check customer types and menu selections. They help the program make decisions based on the user's input.

III. Screenshots of Actual Program Output

This section contains screenshots showing the actual execution of the Campus Café Ordering System.

Screenshot 1: Program Execution

[Insert screenshot showing the program menu and customer information.]

Screenshot 2: Order Summary and Total Payment

[Insert screenshot showing the selected items, prices, and total amount.]

IV. Challenges Encountered and Solutions

While developing the program, we encountered some challenges, particularly in organizing multiple classes and connecting them properly. We also experienced errors involving constructors, inheritance, and method overriding. We solved these problems by reviewing the Java syntax, checking the class names and method declarations, and making sure that the objects were initialized correctly. We also tested the program several times to identify and correct errors.

V. Conclusion

Through this laboratory exercise, we learned how to apply different Object-Oriented Programming concepts in Java. We gained a better understanding of classes, objects, constructors, encapsulation, inheritance, polymorphism, and abstraction. This activity helped improve our programming skills and taught us how to develop a simple but organized ordering system.